

Experiment 10: Implement Expectation & Maximization Algorithm

Aim

To implement the Expectation-Maximization (EM) algorithm in Python for clustering data using a Gaussian Mixture Model (GMM).

Python Program

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
np.random.seed(42)
data1 = np.random.normal(5, 1, 100)
data2 = np.random.normal(10, 1, 100)
X = np.concatenate([data1, data2]).reshape(-1, 1)
model = GaussianMixture(
    n_components=2,
    random_state=42
)
model.fit(X)
labels = model.predict(X)
print("Means of Gaussian
distributions:") print(model.means_)
print("\nVariances:")
print(model.covariances_)
print("\nCluster probabilities:")
```


[illegible]